

Palatable Knowledge is a Seed Without a Forest: A Case Against General Audience

Exposition of Quantum Mechanics

By James Squires

Quantum mechanics; to the average person it's spooky and weird and counterintuitive... to the physicist it's spooky and weird and counterintuitive... and to the mathematician it's math.

My intent by writing this article is to not disparage avid autodidacts or those with the beautiful awe of curiosity, rather, it's to set realistic boundaries that are often shunted by popular narratives of scientific exposition. You may have heard that you only really understand something if you can teach it to a five year old; I won't assert whether this claim is right or not, but I will assert that, in the case of quantum mechanics, this five year old better be damn smart.

In my experience, having written expositional articles at various levels of understanding, from high school level expositions, to undergraduate resources, to high level research materials, I've come to discover an inherent tradeoff between accuracy, specificity, and understandability at the general level. As we make something more understandable or intuitive, this typically means we have to iron out the details and quarks of the subject to give a cleaner picture; this leads to a deceiving and partial picture at best. Here's some descriptions of what people typically say about quantum mechanics:

- Quantum mechanics is the study of how things behave on an incredibly small scale
- Quantum mechanics describes the world probabilistically

- Things are both waves and particles
- Things can be in multiple states simultaneously (Schrodinger's Cat)

While I don't necessarily disagree with any of these claims, there's far more subtlety than meets the eye. When are things in multiple states simultaneously? What does that mean? Why do we say that there's a probability to how things behave; and it turns out things behave far more deterministically on the macroscopic scale... why? I won't attempt to give answers to these questions. The point of this article is to discuss the inherent tradeoff between general understandability and the actual specificity and accuracy of a topic.

Different Levels of Understanding

I'm going to categorize understanding at four layers for the sake of our conversation: novice, apprentice, practitioner, and pioneer.

Novice: the novice is someone just being exposed to the basic concepts of a field or theory. They can be taught some of the basic/simplified equations, and this is where much of the intuition level claims are made. This is typically introductory college or high school level understanding.

Apprentice: the apprentice is being introduced to the theory and motivation behind the equations, they're becoming comfortable with manipulation and usage of the theory. This is typically your undergraduate level.

Practitioner: the practitioner now knows how to reproduce results, following in the footsteps of their forefathers. They can deep dive into topics and they have an intuition for everything the apprentice worked with. This is typically early graduate school level.

Pioneer: this is someone who is not only a practitioner, but a person on the liminal boundaries of the field, leading our understanding to places it hasn't gone before; they are actively engaging in research.

Typically for something like classical mechanics—think Newton's gravity, rocketships, cars, machines, and the world you live in—there's an intuition that accompanies the material. This allows a *novice* approach. However, there's no such intuition for quantum mechanics, as we don't interact with the microscopic world. The best we can do is make analogies, but fundamentally our world isn't the quantum world. So you might ask: "*How do physicists think about quantum mechanics then?*" I can only provide my perspective as someone who falls in between the apprentice and practitioner, that I think about it through the math. Even though math models physics, in the case of quantum mechanics, we want to think of physics through the math. And this math isn't trivial (mathematicians use trivial in a somewhat satirical way; if a mathematician says something isn't trivial, they really mean run for your life because this thing will threaten your sanity).

I assert that the necessary understanding to feel a solid comfortability with undergraduate quantum mechanics stipulates a background in at least an undergraduate mathematics education or something equivalent. But notice, this has completely killed the novice layer that I mentioned. Unlike other branches of physics such as electricity and magnetisms, or Newtonian mechanics, we are missing a layer completely! This is my argument for why quantum mechanics cannot be communicated effectively to the public... the first exposure that a student has to quantum mechanics is through the abstraction of linear algebra and differential equations; what hope is there to explain these things to someone who hasn't taken linear algebra?

I'd like to reiterate this is not meant to be condescending or discouraging; I understand most people don't have the time nor energy, even if they wanted to, to learn differential equations, linear algebra, and probability amongst other things to then understand quantum mechanics. But the unfortunate reality is not all knowledge is meant to be understood without an extraordinary amount of effort. To drive this point home, if I go and ask a math professor "What are the other researchers on your floor doing?" the professor will likely laugh and say they have no idea! Mathematics is arguably more egregious than physics in terms of its opaqueness, but the overlap is blurred when we talk about quantum mechanics or other involved topics. Quite often, not even the researchers in the same department know what most of the other people are doing. At best it will typically be other researchers in their specific subgroup—their research group inside of the physics or math department. This contrasts with a place like the Biology department where the results are typically understandable on the naïve level even if the methodology or context isn't obvious.

A Note on What Exposition Does Accomplish

Many of the people I know got into physics and mathematics from popular expositions such as the ones I'm criticizing. This obviously evidences the strength of general audience expositions: they garner support, appreciation, and admiration for the discipline at hand. When I learn something, I also personally begin my understanding through fragmented pieces of palatable knowledge... intuition should come before formalism, even for the most counter-intuitive ideas. But can you truly say you watched a movie if you only saw the trailer? Can you truly say you read or understand a book if you only read several book reviews? My critique isn't on surface

level knowledge—an essential part of the learning process—but rather the limitations of surface level knowledge.

My Advice

I don't want to leave the reader feeling nihilistic about their pursuit of understanding though, so here's my advice. You still can understand these things to a degree, but there's always a barrier, made of effort and energy, to unlock even higher understanding on any topic. There are so many things I am virtually clueless about in physics and math, and I won't even pretend to know them. If I want to know these things, I have to engage with them in a way that takes time and effort. I sometimes compare this to fast food. The general audience information conveyed, especially on places like YouTube shorts or TikTok etc. are fast-food information. They don't enrich your perspective, rather they're morsels of knowledge extracted and processed to be easily palatable. But we know that people understand ideas not in isolation, but rather through their deep interconnection between one another. If we plant a lone tree, it will wither and die. But if we plant a forest, it will grow and thrive. Understanding is fostered through interwoven ideas. The understanding of quantum mechanics and many other areas of human knowledge can only fully flourish as an enriched network of ideas that take years of mastery and depth. To quote Aristotle, *The Whole is greater than the Part.*